LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	34
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Remote indicating compasses: Essential components and principle of operation of flux detector
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Remote indicating compasses: - Essential components and principle of operation of flux detector.
5. Specific Objectives	15.1.12.3 Trainee will be able to explain Remote indicating compass system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB schematics; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary limitation of a direct reading magnetic compass?
(b) Name two essential components of a remote indicating compass system.
(c) How does a flux detector differ from a standard magnetic compass?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Essential components of remote indicating compass • Function of flux detector in system • Magnetic field generation in flux detector • Induction coil operation and signal generation • Signal transmission to remote indicator	L & Ds	PPT, IB	15 min
(b)	• Principle of operation of flux detector • Interaction between permanent magnet and coil • Conversion of magnetic flux to electrical signal • Role of soft iron core in flux concentration • Output signal characteristics for indicator	L & Ds	PPT, PDF	20 min
(c)	• System integration with aircraft electrical power • Error correction mechanisms in remote systems • Maintenance checks for flux detector assembly • Troubleshooting common flux detector failures • Safety precautions during system inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the flux detector converts magnetic flux into an electrical signal.
(b) Describe the role of the induction coil in the remote indicating compass system.
(c) Identify three essential components required for a functional remote indicating compass.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the flux detector principle and the gyro magnetic compass principle to be discussed in LP 35.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation of gyro magnetic compass system.	Refer to the master syllabus reference list	LP 35

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min